

PAPER_954: E(t) Linewidth Modulation with Sign-Flip Dynamics

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Abstract

We derive the $E(t)$ linewidth modulation function with sign-flip dynamics for astrophysical jets. The time-domain response $E(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t)$ exhibits sign flips at $t_{\text{flip}} = \pi/(2\omega_{\text{SCm}})$, driving extra-gravitational responses in blazar jets. Narrow linewidths produce sharper sign flips with tighter jet collimation; broad linewidths damp the oscillation before the first flip.

1. E(t) Function

$$E(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} \cdot t) \cdot \exp(-\Gamma \cdot t)$$

2. Sign-Flip Time

$$t_{\text{flip}} = \frac{\pi}{2\omega_{\text{SCm}}} \approx 0.064 \text{ ps}$$

3. Regime Dependence

Γ (THz)	Flips in 5 ps	Jet Behavior
0.01	Many	Ultra-tight collimation
0.10	Several	Optimal modulation
1.00	Few/None	Damped, diffuse wind

4. Source Data

- **File:** blazar_jet_phonon.py
- **Session:** 214
- **CP4 Class:** EtLinewidthModulationCalc (#538)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)